

10. Adnexal Mass Management: Observation, Surgery, Torsion, Adolescents, Pregnancy

Companion reading handout for simultaneous listening.

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Observation is not neglect.

And surgery is not automatically safety.

That is the central tension in adnexal mass management. A resident who operates on every mass will harm patients with benign disease. A resident who reassures every patient with a cyst will miss the moment when malignancy, torsion, infection, growth, or symptoms should change the plan.

So management is not "operate or do not operate."

Management is a ladder.

The ladder is: active observation, repeat imaging, benign minimally invasive surgery with ovarian preservation, gynecologic oncology referral, emergency torsion surgery, and special modification for adolescents and pregnancy.

Lecture 9 taught the first step: understand the patient context and ultrasound architecture. This lecture asks what to do with that information.

Observation

Observation is appropriate when ultrasound morphology suggests benign disease, or when morphology is less certain but there is a compelling reason to avoid surgery. For an asymptomatic patient, observation may be justified when CA 125 is normal and transvaginal ultrasound does not show suspicious features.

Simple cysts are the classic example. With rare exception, simple cysts up to ten centimeters on transvaginal ultrasound performed by experienced ultrasonographers are likely benign and may be monitored with repeat imaging without surgical intervention, even in postmenopausal patients.

That sentence deserves attention because it protects patients from unnecessary operations.

The key words are simple, expertly characterized, and monitored.

Simple means thin smooth walls, no solid components, no septations, and no internal blood flow. Expertly characterized means the ultrasound is good enough to trust the morphology. Monitored means there is a follow-up plan.

So the resident reflex is this: observation is active surveillance, not abandonment.

The follow-up interval is not perfectly defined. ACOG notes that repeat ultrasound imaging is recommended whenever the diagnosis is uncertain and cancer remains in the differential. In one study, masses that were monitored and later diagnosed as malignancies all showed growth by seven months. Some experts limit observation of stable masses without solid components to one year, and stable masses with solid components to two years.

These are not magic cutoffs. They are risk-control concepts. If the mass is truly benign-appearing and stable, observation can prevent unnecessary morbidity. If the mass is uncertain, repeat imaging is a way to let behavior declare itself while still maintaining vigilance.

Surgery

Surgical intervention is warranted for symptomatic masses or suspected malignancy based on imaging, serum markers, or both.

Symptoms matter. Pain, pressure, growth, concern for torsion, persistent mass, or complications can justify surgery even when malignancy is not the main concern.

But surgery risk also matters. Some patients who would otherwise be considered for surgery have very high perioperative risk because of advanced age or major comorbidity. In those patients, repeat imaging may be safer than immediate operation, although the ideal interval is not firmly established.

For presumed benign adnexal masses, minimally invasive surgery is preferred when feasible. Laparoscopy generally offers shorter hospitalization, less pain, less perioperative morbidity, and faster recovery than laparotomy for clinically benign masses.

ACOG describes three randomized trials including three hundred ninety four patients comparing laparoscopy with laparotomy for clinically benign pelvic masses. Conversion to laparotomy occurred only for endoscopic suspicion of cancer, with conversion rates from zero to one point five percent.

That evidence supports a practical rule.

If the mass is presumed benign and surgery is needed, choose a route that reduces morbidity, but stay prepared to change course if cancer becomes plausible.

Fertility preservation is central in adolescents and premenopausal women who have not completed childbearing. Even with large ovarian cysts of ten centimeters or more, it may be possible to save normal ovarian tissue and remove the cyst laparoscopically.

The goal is not just removing the mass. The goal is removing the problem while preserving endocrine and reproductive function when oncologically safe.

Malignancy-risk masses

If the mass is suspicious and surgery is needed, the question is not only route. It is surgeon, facility, staging, and timing of referral.

Referral to, or consultation with, a gynecologic oncologist is recommended for postmenopausal patients with an adnexal mass plus elevated CA 125, suspicious ultrasound findings, ascites, a nodular or fixed pelvic mass, or evidence of abdominal or distant metastasis.

For premenopausal patients, referral or consultation is recommended with very elevated CA 125, suspicious ultrasound findings, ascites, a nodular or fixed pelvic mass, or evidence of abdominal or distant metastasis.

Referral is also recommended when a formal risk assessment test is elevated, such as the multivariate index assay, risk of malignancy index, Risk of Ovarian Malignancy Algorithm, or an International Ovarian Tumor Analysis score.

Why does this matter?

Because if ovarian cancer is present, the first operation is not just removal of a mass. It is staging, cytology, inspection, biopsies when indicated, and sometimes debulking. ACOG emphasizes that women treated by physicians with advanced training in ovarian cancer have improved survival. That improvement reflects proper staging and aggressive debulking of advanced disease, which is present in seventy five to eighty percent of women with ovarian cancer.

So the resident reflex is this: refer before the wrong operation, not after it.

If a suspicious or persistent complex mass requires surgery, it should be managed by a physician trained to stage and debulk ovarian cancer, in a facility with appropriate support, including pathology support such as frozen section when needed.

Aspiration

Aspiration can seem tempting. It sounds less invasive than surgery. It sounds like a way to make a cyst go away. But diagnostic aspiration is generally contraindicated when cancer is possible.

There are two main reasons.

First, cytology sensitivity is limited. ACOG cites sensitivity only around fifty to seventy four percent. A negative aspirate cannot safely rule out cancer.

Second, if the mass is malignant, aspiration can spill and seed malignant cells into the peritoneal cavity. That can create more advanced stage disease at diagnosis and potentially worsen prognosis.

Even when the mass is a benign simple cyst, aspiration often is not durable therapy. In one series, recurrence by six months was forty four percent in premenopausal women and twenty five percent in postmenopausal women. Another series reported recurrence around thirty nine percent.

There are exceptions, and the exceptions prove the rule.

Drainage may be appropriate for tubo-ovarian abscess in selected cases, although antibiotic therapy is first-line. Aspiration may also be used when there is clinical and radiographic evidence of advanced ovarian cancer, the patient is medically unfit for surgery, and malignant cytology would allow neoadjuvant chemotherapy.

But for the ordinary adnexal cyst where cancer remains possible, the reflex is clear.

Do not aspirate a mass when cancer is still possible.

Torsion

Torsion is different from the slow risk ladder. It is an emergency branch.

The concern is twisting of the adnexa with compromised vascular flow. The classic presentation is sudden unilateral pelvic pain, often with nausea and vomiting, but the clinical picture can vary. Once torsion is suspected strongly enough, management is surgical.

The modern principle is ovarian conservation.

For patients who desire fertility, adnexal torsion should be managed by detorsion and cystectomy for identified ovarian pathology when appropriate. Conservation should be prioritized because the residual ovary usually regains perfusion and remains viable.

This is one of the most important memory points in the lecture.

Dark does not mean dead.

At surgery, a torsed ovary may look blue, black, congested, or ischemic. That appearance does not prove the ovary cannot recover. ACOG notes that ovarian function is preserved in more than ninety percent of cases three months after intervention, despite evidence of ischemia or necrosis at exploration.

So do not remove a torsed ovary just because it looks frightening. Detorse. Treat the cyst if needed. Preserve ovarian tissue when safe.

Ovarian fixation remains controversial, but it may be considered in recurrent torsion.

Adolescents

Adolescent adnexal mass management uses the same broad evaluation as premenopausal patients, but the emphasis on ovarian conservation becomes even stronger.

Many adolescent adnexal masses are benign and can be managed expectantly. The evaluation should include menstrual history and a confidential inquiry about sexual activity. For young, virginal, or prepubertal adolescents, transabdominal ultrasound rather than transvaginal ultrasound is recommended.

If a germ cell tumor is possible, obtain alpha fetoprotein, beta hCG, and LDH. Germ cell tumors are the most common ovarian malignancies in children and adolescents.

Surgical indications include suspected malignancy, torsion, persistent mass, and acute abdominal pain.

When surgery is needed, the operative plan should protect future fertility whenever oncologically safe. Benign masses may be treated with cystectomy rather than removal of the ovary when possible. If malignancy is identified, correct staging still matters; ovarian conservation must be balanced against oncologic safety.

The adolescent reflex is this: preserve the ovary by default, but do not under-stage cancer.

Mature teratomas and endometriomas

Mature teratomas, often called dermoids, are usually benign, but they are not always ignored. Surgery is warranted if they are large, symptomatic, growing on serial imaging, or suspicious for malignancy. If managed expectantly, surveillance is needed.

ACOG cites a series of two hundred eighty nine women who chose expectant management for ultrasound-diagnosed mature teratomas. Twenty six percent eventually underwent surgery. Those who failed expectant management were more likely to have larger or more rapidly enlarging cysts.

So the resident should not think of expectant management as "the dermoid disappeared from our responsibility." If you observe it, you follow it.

Endometriomas require even more fertility-aware thinking.

Surgery is warranted if an endometrioma is large, symptomatic, growing, or suspicious for malignancy. But surgical excision can adversely affect ovarian reserve. ACOG notes that endometriomas of five centimeters or more have been associated

with lower ovarian follicle density, and that asymptomatic endometriomas do not require intervention solely for infertility.

That is a subtle but crucial point.

The presence of an endometrioma does not automatically mean surgery improves fertility. In an asymptomatic infertility patient, surgery may remove disease but also reduce ovarian reserve. If surgery is required, conserve as much ovarian tissue as possible.

The endometrioma reflex is this: do not trade ovarian reserve for a surgery that has no clear indication.

Pregnancy

Most adnexal masses in pregnancy are low risk and may be managed expectantly.

ACOG reports that fifty one to ninety two percent of adnexal masses resolve during pregnancy. Persistence is more likely when the mass is greater than five centimeters or has complex morphology on ultrasound. Acute complications are reported in less than two percent. Malignancy among persistent masses is uncommon but real, about one point two to six point eight percent.

Pregnancy changes both imaging and markers.

Depending on gestational age, abdominal ultrasound may be needed in addition to transvaginal ultrasound because the ovaries may move out of the pelvis later in pregnancy.

If additional imaging is needed, MRI is preferred because it can evaluate deep soft tissue without ionizing radiation.

CA 125 is difficult to interpret in pregnancy. It is elevated especially in the first trimester. ACOG cites a first-trimester range from seven to two hundred fifty one units per milliliter, with levels decreasing thereafter. Low-level elevations in pregnancy are usually not associated with malignancy.

So do not let a mildly elevated CA 125 in early pregnancy create a false cancer diagnosis.

Intervention in pregnancy is reserved for symptoms, complications, suspicion of malignancy, or persistence with concerning features. If surgery is warranted, both laparoscopy and laparotomy may be considered depending on gestational age, urgency, mass features, and expertise. ACOG notes data supporting the relative safety and efficacy of laparoscopic management of persistent adnexal masses in the second trimester.

Apply the ladder through cases

Case one.

A postmenopausal patient has a simple unilocular cyst less than ten centimeters, thin smooth wall, no solid component, no septations, no internal flow, no suspicious symptoms, and a normal CA 125.

This is not automatically an operation. If the ultrasound was expertly performed and the whole picture is reassuring, repeat imaging surveillance is reasonable.

Case two.

A reproductive-aged patient has a symptomatic presumed benign cyst. Pain is persistent, malignancy features are absent, and fertility matters.

This patient may need surgery, but the surgical goal is cystectomy and ovarian preservation when safe, preferably by a minimally invasive route.

Case three.

A patient has a persistent complex mass with papillary projections, ascites, and elevated risk markers. Surgery is being discussed.

This is not the moment for casual cystectomy by the nearest available surgeon. This is the moment for gynecologic oncology referral or consultation before the operation.

Case four.

A patient asks for aspiration because she wants a quick office solution.

If cancer is possible, the answer is no. Aspiration can miss malignancy, spill malignant cells, and often fails as durable therapy even for benign cysts.

Case five.

At laparoscopy for torsion, the ovary looks dark and ischemic.

Do not translate dark into dead. Detorse, preserve the adnexa when safe, and perform cystectomy for identified pathology when appropriate.

The final synthesis

Adnexal mass management is a ladder, not a binary.

Observe low-risk morphology actively.

Repeat imaging when uncertainty remains.

Operate minimally invasively for presumed benign symptomatic disease while preserving ovarian tissue when safe.

Refer malignancy-risk masses before surgery.

Avoid aspiration when cancer is possible.

Treat torsion urgently, and remember that a dark ovary can recover.

In adolescents, prioritize ovarian conservation without ignoring malignancy. In pregnancy, expectant management is often appropriate, MRI is preferred if additional imaging is needed, and CA 125 is hard to interpret.

The final resident reflex is this.

Choose the step on the ladder that prevents the real risk: cancer undertreatment, surgical overtreatment, ovarian loss, torsion loss, pregnancy harm, or missed follow-up.